University of Prishtina

Faculty of Electrical and Computer Engineering

First Intrusion Report

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Flags and Services
Figure 1. Flags status in IScorE
Active System and Services
Public services exposed (Team 9):
Domain Name
Public IP
Port(s)
ad.team9.isucdc.com
33.96.50.10
3389, 389, 636
tickets.team9.isucdc.com
33.96.50.20
3389, 5000
mgmt.team9.isucdc.com
33.96.50.30
22, 5000
admin.team9.isucdc.com
33.96.50.40
3389, 80
www.team9.isucdc.com
33.96.50.50
22, 80
Logs explaining
Figure 2. Logs from Elastic Stack
After reviewing authentication logs from Windows systems (ad.team9.isucdc.com, tickets.team9.isucdc.com, admin.team9.isucdc.com), we observed a mixture of interactive and network logon events that align with expected system behavior. System processes and service accounts generated routine logons, and no suspicious user creation, group membership changes, or anomalous privilege assignments were identified in the reviewed window.
Figure 3. Authentication logs
Correlation across hosts shows routine network activity on the AD and application servers, including service lookups and internal service-to-service communications. Any external authentication attempts were cross-checked against endpoint and firewall telemetry to confirm whether the attempts progressed beyond initial probing. In the reviewed sample, activity did not progress to confirmed unauthorized access.
Figure 4. Discovery of the logs on the Servers
During the attack phase the attackers tried to enter via ssh, and this is shown from the figure below
Conclusion

]Flags and Services

This report presents an analysis of Team 9's service security posture during the reporting window, with emphasis on service availability, monitoring observations, and evidence of unauthorized access. Overall, most services remained stable and responsive; however, we identified and confirmed compromises affecting both the WWW (/etc/) Blue Flag and the WWW (/root/) Red Flag, each shown as Claimed during this period.

Upon detection, we treated these events as confirmed intrusions against the web service and immediately initiated containment and remediation actions. We reviewed relevant host and service logs, validated file integrity in and around the /etc/ and /root/ paths associated with the WWW service, and reduced exposure by applying configuration hardening and stricter access controls (including firewall/ACL adjustments where applicable). Following these actions, we continued monitoring for repeat indicators (scanning activity, authentication failures, abnormal requests) and verified normal service behavior post-mitigation. Evidence of the compromise and our response actions is included in the supporting artifacts/screenshots and the referenced log excerpts within this report.

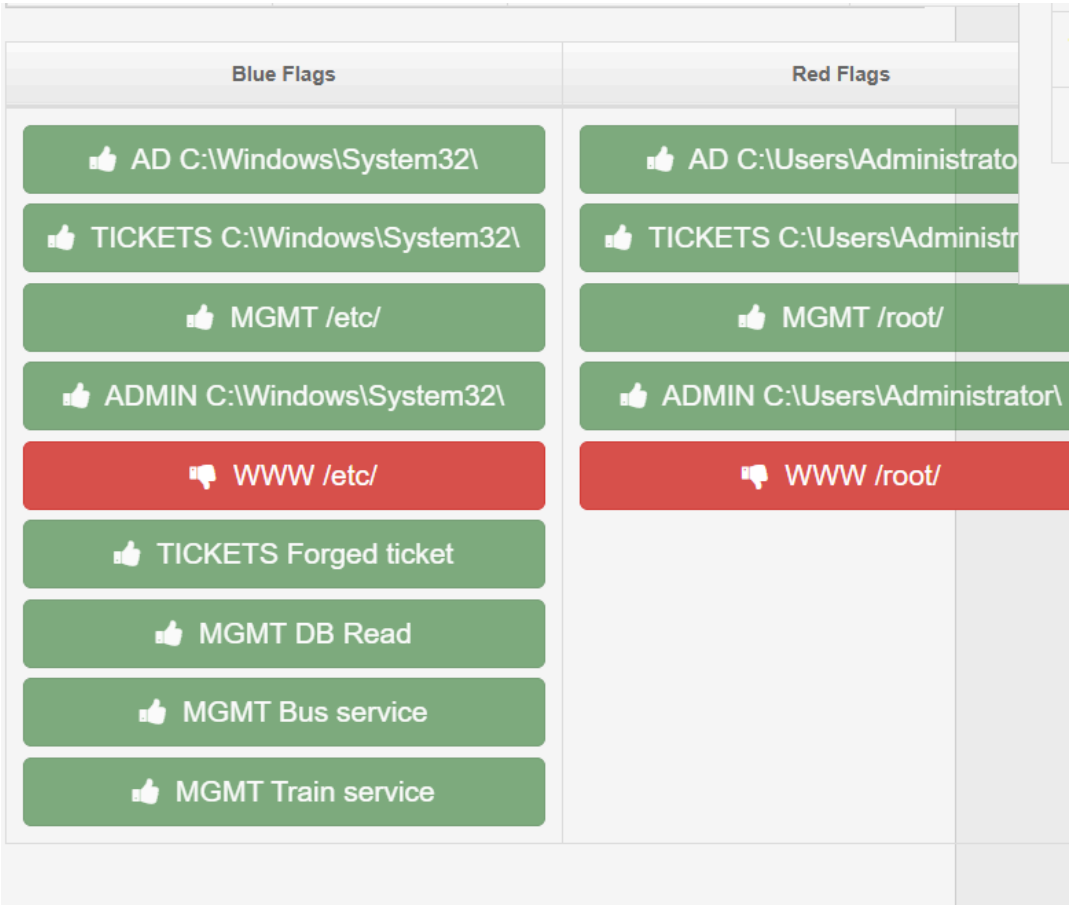


Figure 1. Flags status in IScorE

Active System and Services

The analysis confirms that the system and services are currently active and operational. Through comprehensive monitoring and assessment, no confirmed indicators of unauthorized entry were detected during the reporting window. This suggests that the implemented security measures have effectively safeguarded the system's infrastructure from external threats and unauthorized access.

Public services exposed (Team 9):

Table1: Team 9 Public Services Exposure

Domain Name	Public IP	Port(s)
ad.team9.isucdc.com	33.96.50.10	3389, 389, 636
tickets.team9.isucdc.com	33.96.50.20	3389, 5000
mgmt.team9.isucdc.com	33.96.50.30	22, 5000
admin.team9.isucdc.com	33.96.50.40	3389, 80
www.team9.isucdc.com	33.96.50.50	22, 80

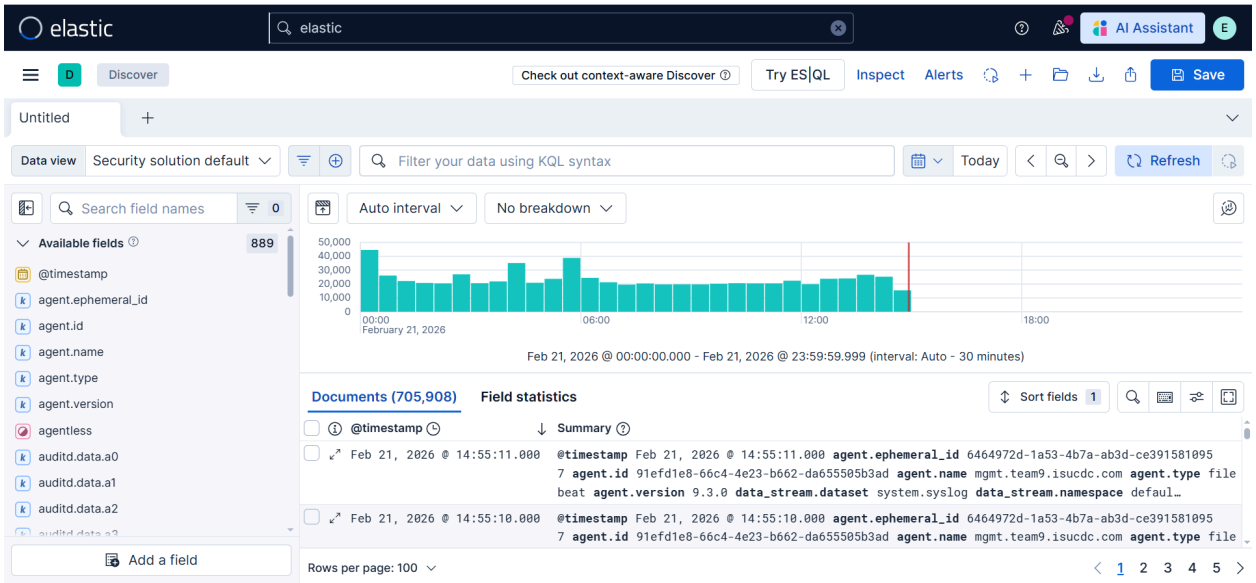
Logs explaining

This section summarizes key observations from centralized logging (Elastic Stack), IDS/IPS alerts (Suricata), firewall events (pfSense), Windows authentication telemetry (AD/Tickets/Admin), Linux authentication telemetry (MGMT/WWW), and web/application access logs. Screens hots or exports referenced as figures should be attached in the final submission.

During the reporting window, a small number of authentication events were observed that are consistent with normal administrative activity (e.g., green team validation) and general Internet background noise (scans, login probes). No evidence of successful privilege escalation, persistence, or flag access was identified.

On the management server (mgmt.team9.isucdc.com), SSH authentication logs recorded legitimate access associated with administrative operations. In the same timeframe, repeated failed login attempts were also observed from external sources. The timing and pattern of failures (multiple attempts within seconds/minutes against different usernames) is consistent with reconnaissance or password spraying behavior. These attempts were monitored and

correlated with firewall and IDS/IPS telemetry to determine whether any follow-on actions occurred.



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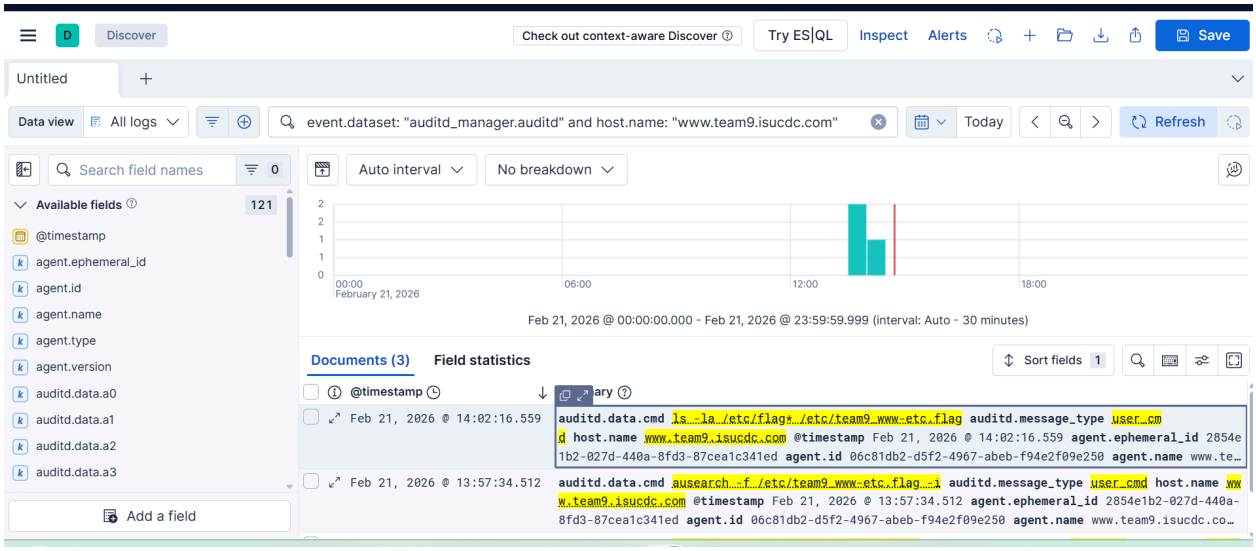
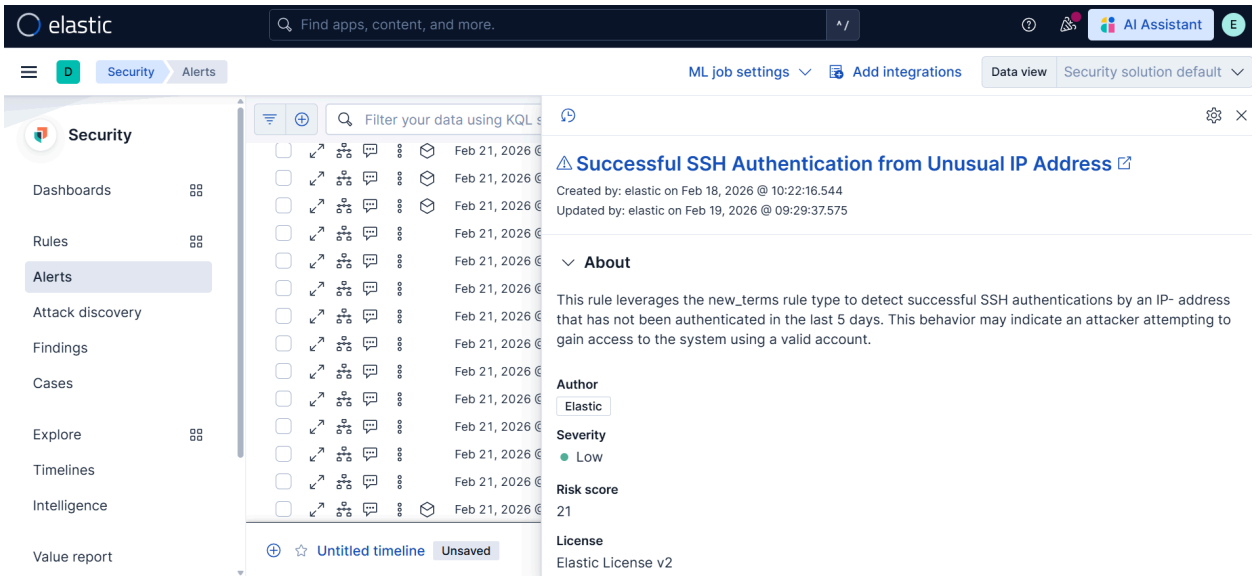


Figure 4. Discovery of the logs on the Servers

During the attack phase the attackers tried to enter via ssh, and this is shown from the figure below



After not successfully logging into the www machine they tried password spraying attacks against the host.

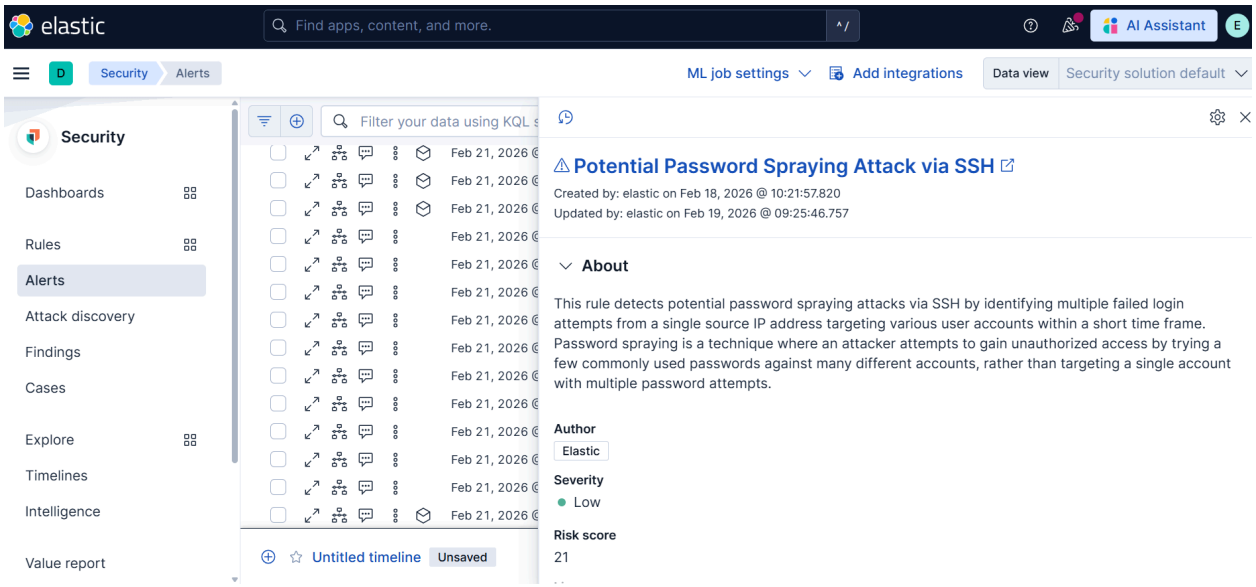


Figure 6.Highlights pfSense firewall logs ingested into Elastic.

Observed blocked events include unsolicited inbound connections and traffic matching configured deny rules. Traffic was assessed based on source/destination IPs, ports, and rule identifiers to confirm that protective controls were working as intended.

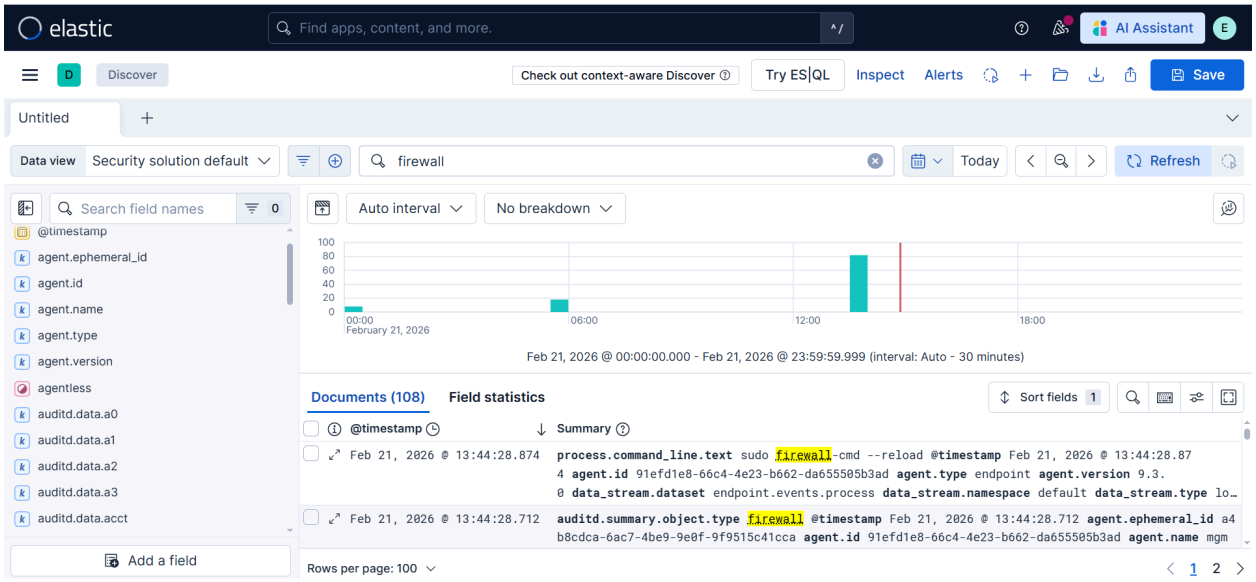


Figure 6. Firewall logs shown in Elastic Stack

As the stack shows, after successfully entering the WWW machine, the attackers used a malicious script named `pscript.sh`. The activity was logged under the `cdc` account, and a cron job was used to trigger the script for persistence. During the response and remediation phase, we identified and removed an unauthorized user account that had been created on the system.

Conclusion

Based on the reviewed telemetry from Elastic Stack, pfSense firewall logs, Windows and Linux authentication logs, and host-level observations, Team 9 systems remained largely operational during the reporting window; however, the analysis confirms **malicious activity and a successful compromise on the WWW server**.

Evidence indicates that attackers initially conducted **reconnaissance and password spraying attempts**, particularly against exposed services (including SSH). While several attempts were blocked or unsuccessful, the logs and host artifacts show that the attackers **eventually gained access to the WWW machine**. After access was obtained, they executed a **malicious script (`pscript.sh`)**, leveraged the `cdc` **account**, and established **persistence via a cron job**.

In addition, the competition flag status and supporting evidence confirm that both the **WWW (/etc/) Blue Flag** and the **WWW (/root/) Red Flag** were **claimed/compromised** during the reporting window. This indicates that the intrusion progressed beyond probing and resulted in unauthorized access to sensitive locations on the host.

Following detection, containment and remediation actions were initiated, including log review and correlation, verification of affected paths, removal/analysis of malicious artifacts, and hardening measures (e.g., stricter access controls and firewall/ACL tuning). Post-mitigation monitoring was continued to identify repeat activity and verify service stability.

Overall, the report concludes that **Team 9 experienced a confirmed intrusion affecting the WWW service**, while other monitored systems primarily showed routine or unsuccessful probing activity during the same period. Continued monitoring, improved credential hygiene, stronger authentication controls, and tighter alerting/tuning remain essential to reduce the likelihood and impact of similar attacks in future competition windows.